
AI-102 Natural Language Processing (NLP)

Dr. Jimmy Ardiansyah



EMERGING &
COLLABORATIVE STUDIES





| Syntax and Parsing

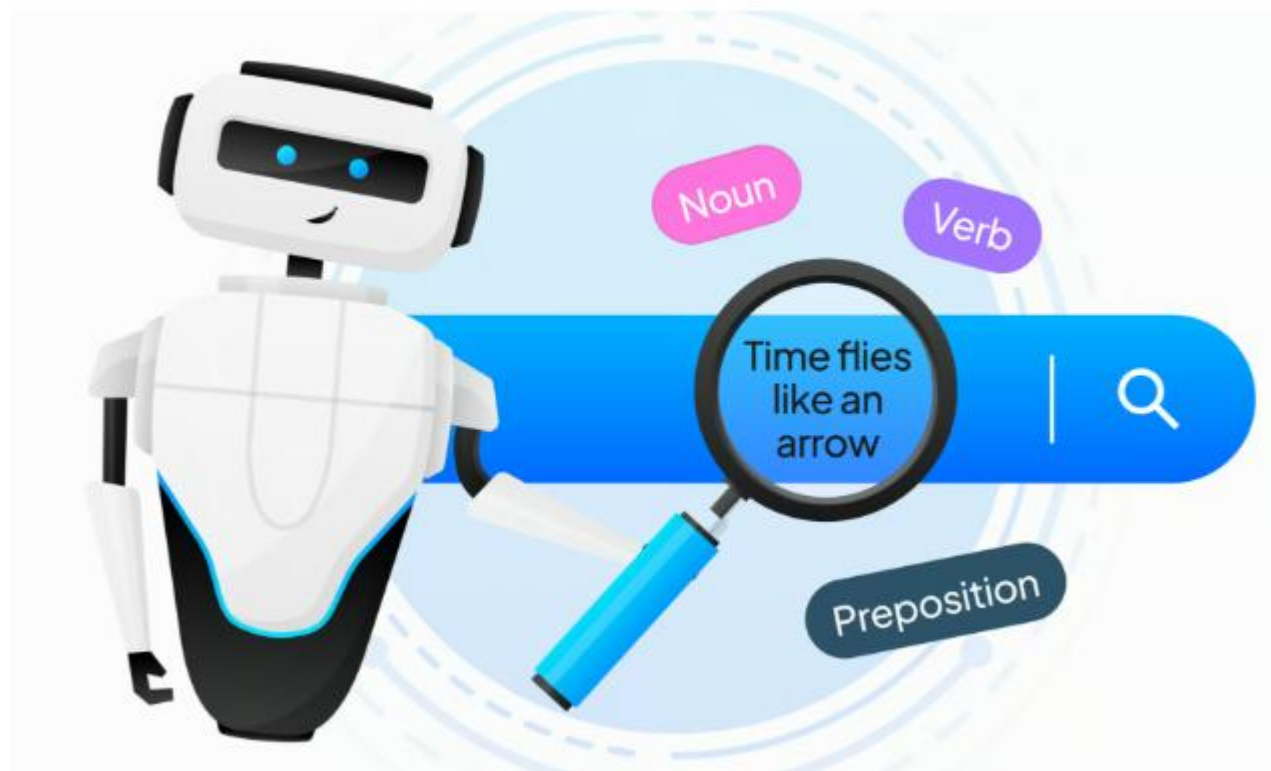
Part of Speech (POS) Tagging

Part-of-Speech (POS) Tagging is the process of assigning grammatical categories (noun, verb, adjective, etc.) to each word in a sentence.

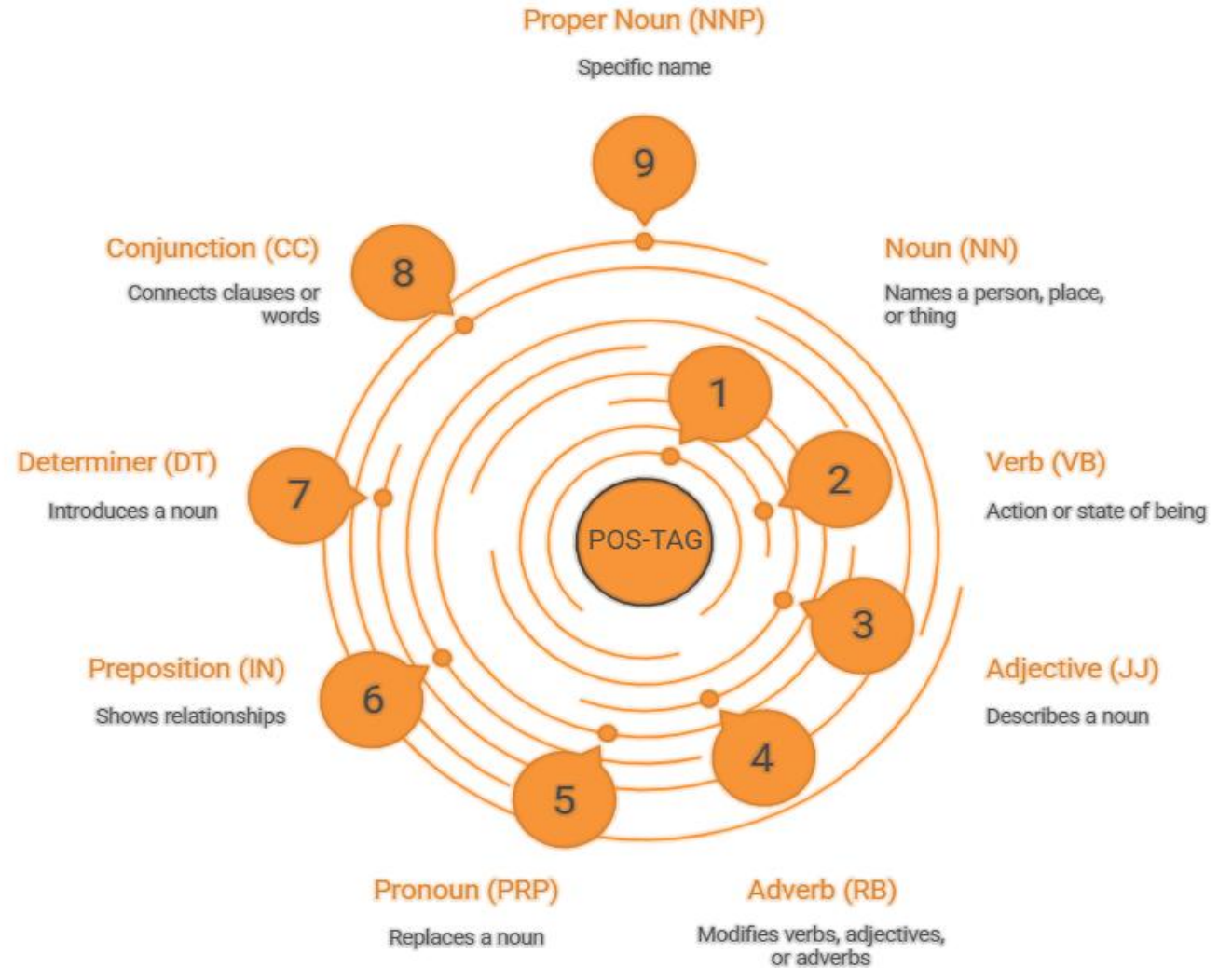
Purpose: Helps computers understand the syntactic structure and function of words. POS tagging is a foundation for advanced NLP tasks like parsing, sentiment analysis, and machine translation.

Example:

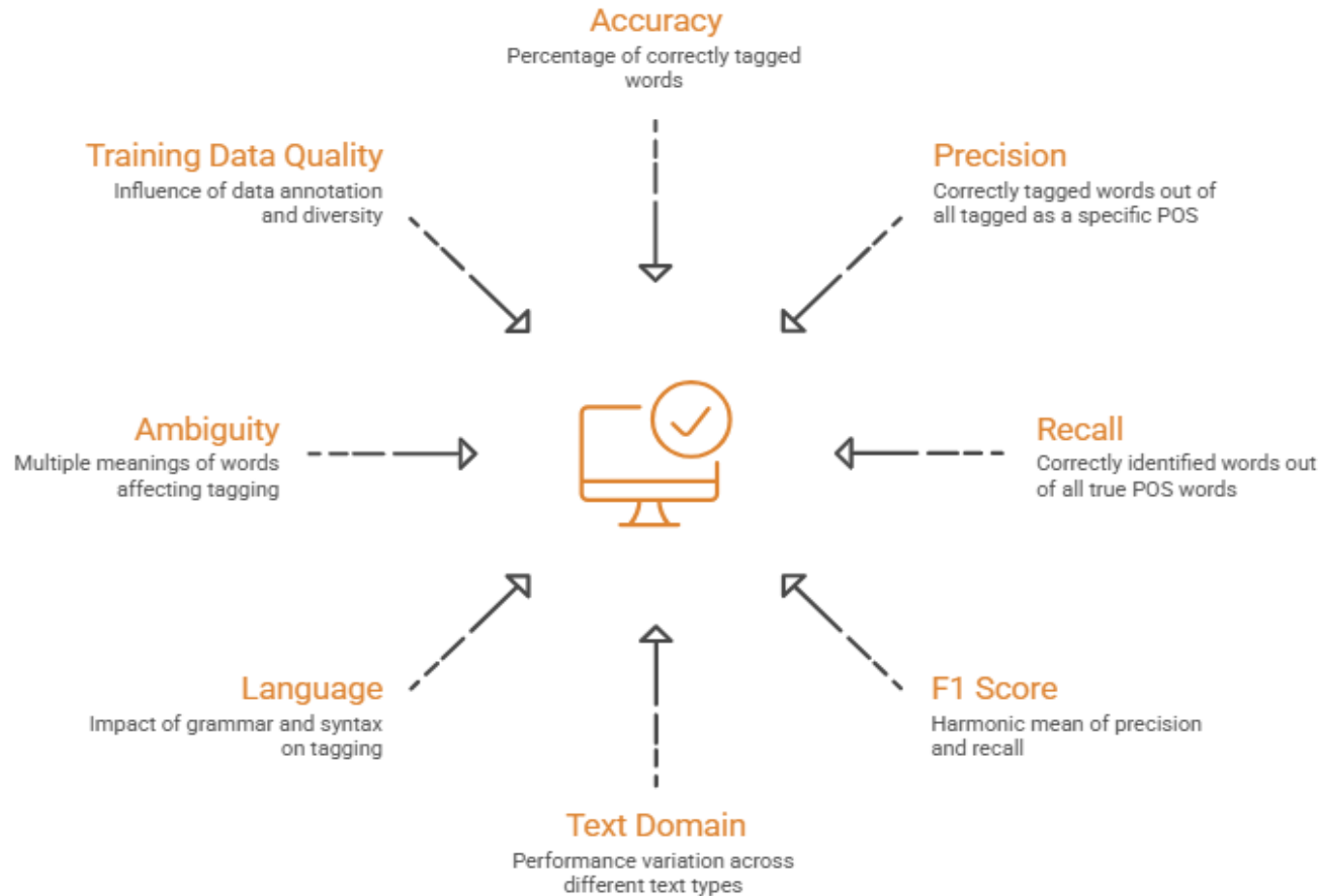
Sentence: “*Natural Language Processing with Python is fascinating.*” → [('Natural', JJ), ('Language', NN), ('Processing', NN), ('with', IN), ('Python', NNP), ('is', VBZ), ('fascinating', VBG)]



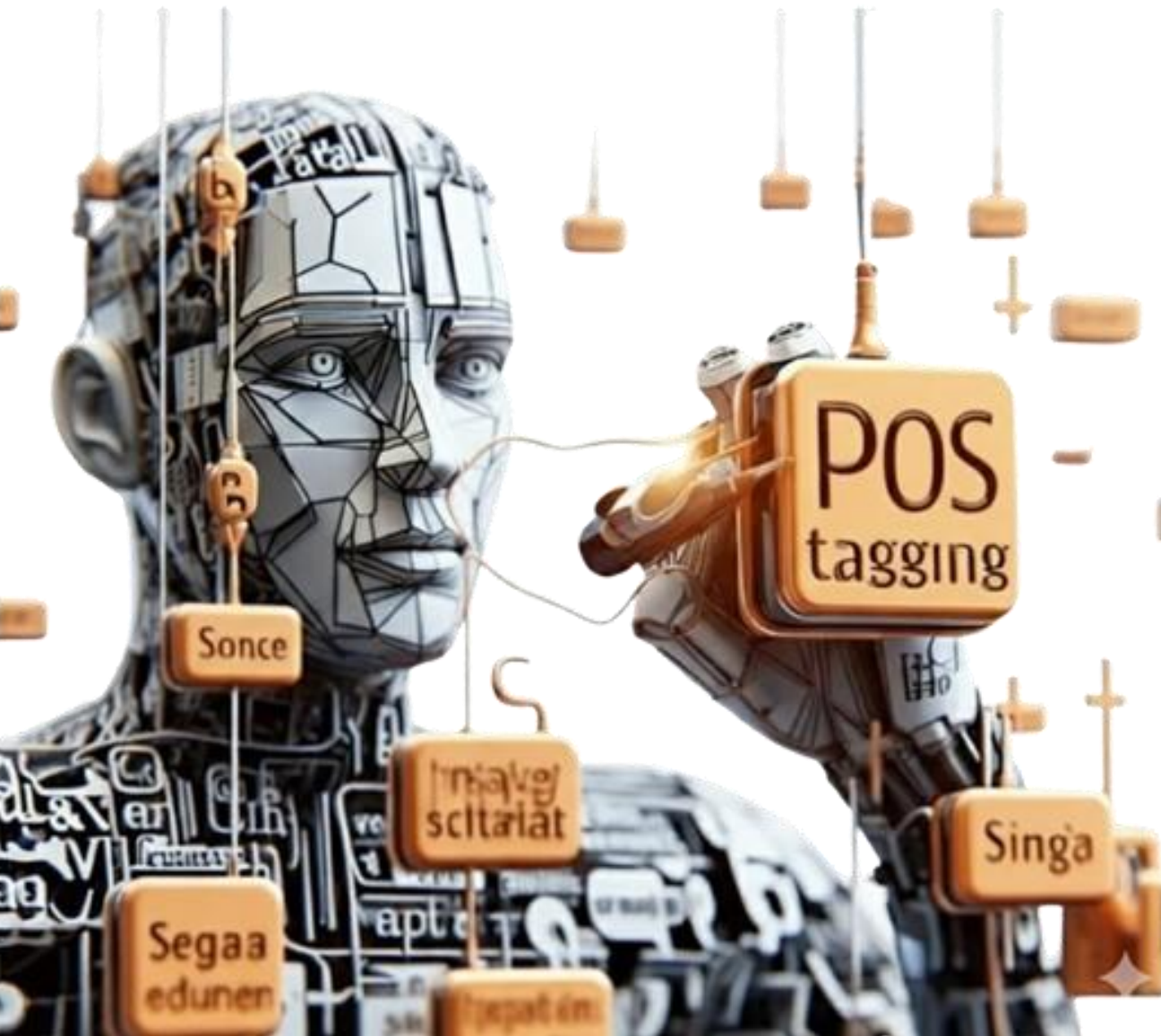
Understanding POS Categories



Evaluate Part of Speech (POS) Tagger



Applications of Part of Speech (POS) Tagging



- **Parsing:** Builds grammar trees; defines sentence structure.
- **Named Entity Recognition (NER):** Distinguishes nouns for names, locations, etc.
- **Sentiment Analysis:** Detects emotion-bearing words (e.g., adjectives).
- **Information Extraction:** Finds key subjects, actions, and objects.
- **Machine Translation:** Maintains grammatical consistency across languages.
- **Text-to-Speech (TTS):** Affects intonation and stress patterns.
- **Grammar Checking:** Detects tense errors, agreement issues, etc.

What's Named Entity Recognition (NER)

(NLP) technique that identifies and categorizes key information (named entities) in text into pre-defined groups such as **Person, Organization, Location, Date/Time, Money, Percent, Quantity**, etc.

Why use NER?

- **Question answering:** find the exact entity asked for.
- **Information retrieval & search:** rank or filter by important entities.
- **Content tagging & analytics:** auto-label articles by people, places, orgs.
- **Customer support:** detect products, issues, and locations in tickets.



Application of Named Entity Recognition (NER).



What's Dependency Parsing

Dependency parsing is a natural language processing (NLP) technique that analyzes the grammatical structure of a sentence by identifying **direct relationships between "head" words and the "dependent" words that modify them**. The output is a tree-like structure (a dependency tree) that maps out these word-to-word connections and their specific grammatical role

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Key Concepts

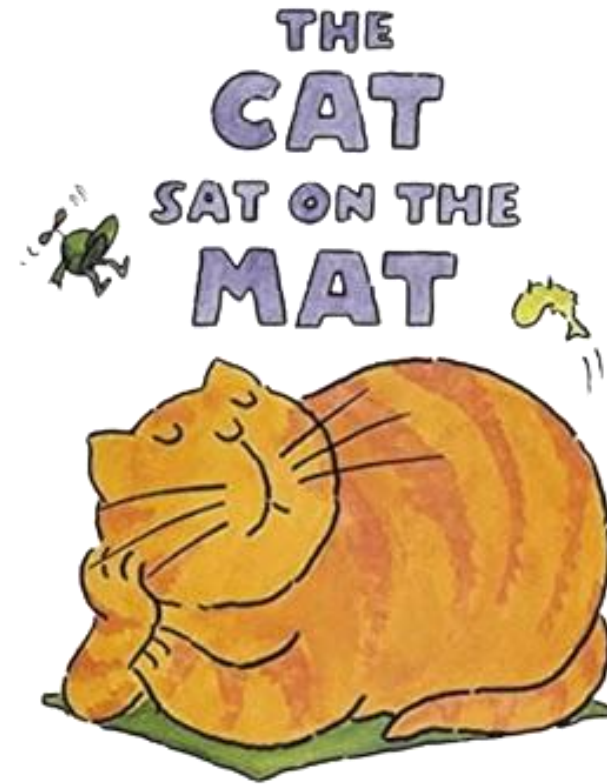
- **Head (Governor):** The central or governing word in a relationship (e.g., the main verb of the sentence).
- **Dependent (Modifier):** The word that relies on the head for its full meaning or role in the sentence (e.g., subjects, objects, or adjectives).
- **Dependency Labels:** Each connection (arc) in the dependency tree is labeled to describe the exact nature of the relationship, such as *nsubj* (nominal subject), *dobj* (direct object), *amod* (adjectival modifier), or *prep* (prepositional modifier).
- **Root:** The top-most word in the tree, which is usually the main verb of the sentence and has no incoming arcs.

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Example

For the sentence, "The fat cat sat on the mat":

- *sat* is the **root** and the main head.
- *cat* is a **dependent** of *sat* with the label *nsubj* (nominal subject).
- *The* (modifying cat) is a **dependent** of *cat* with the label *det* (determiner).
- *on* is a **dependent** of *sat* (preposition).
- *mat* is a **dependent** of *on* (object of the preposition)





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